

Sample Output for Frozen Lake Q-Learning

Environment: FrozenLake-v1

Total States: 16 states

Each state represents: the current grid position of the agent.

Actions:

0 → Left
1 → Down
2 → Right
3 → Up

Q-Learning Equation:

$$Q(s,a) = Q(s,a) + \alpha[\text{Reward} + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Intuition:

New estimate = old estimate +

correction from:

- reward
- future best value

Sample run 1:

For,

alpha = 0.1 # learning rate
gamma = 0.99 # discount factor
epsilon = 0.1 # exploration probability
total episodes = 10000 # one complete game

```
Final Q-Table:
```

```
[[0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]]
```

```
Total Reward:
```

```
0
```

Sample run 2:

For,

```
alpha = 0.1      # learning rate
```

```
gamma = 0.99     # discount factor
```

```
epsilon = 0.3    # exploration probability
```

```
total episodes = 100000 # one complete game
```

Final Q-Table:

```
[[0.94148015 0.95099005 0.95099005 0.94148015]
 [0.94148015 0.          0.96059601 0.95099005]
 [0.95099005 0.970299   0.95099005 0.96059601]
 [0.96059601 0.          0.94937313 0.94805152]
 [0.95099005 0.96059601 0.          0.94148015]
 [0.          0.          0.          0.          ]
 [0.          0.9801     0.          0.96059601]
 [0.          0.          0.          0.          ]
 [0.96059601 0.          0.970299   0.95099005]
 [0.96059601 0.9801     0.9801     0.          ]
 [0.970299   0.99       0.          0.970299   ]
 [0.          0.          0.          0.          ]
 [0.          0.          0.          0.          ]
 [0.          0.9801     0.99       0.970299   ]
 [0.9801     0.99       1.          0.9801     ]
 [0.          0.          0.          0.          ]]
```

Total Reward:

63502

Note:

Even with the same hyperparameters and number of episodes, different runs may produce different results due to random exploration and stochastic learning behavior during training.

(The environment here is deterministic but learning process is still random)

Important Observations:

Initially:

the Q-table remained mostly zeros because the agent rarely reached the goal.

After increasing:

- number of episodes,
- and exploration rate (epsilon),

the agent successfully discovered rewarding paths and learned useful state-action values.